==
VECTOR CALCULUS SIMULATION REPORT
Dimensional Collapse in Boolean Hypercubes
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Date: February 05, 2026
Experiment: Information Flow Field & Differential Collapse Analysis

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MODEL DESCRIPTION
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This report analyzes the vector calculus framework for dimensional collapse in Boolean function minimization. Two main models are implemented:

1. Information Flow Field (PDE)
• Vector field: $F(n,\rho,D) = \nabla\Psi$
• Divergence analysis: $\nabla\cdot F$ to detect phase transitions
• Critical points where $\nabla\cdot F \approx 0$

2. Differential Collapse Equation (ODE)
• Evolution: $d\Psi/dn = \Gamma - \Gamma_{crit} - 2\alpha[I_{avg} - I_{sat}]\cdot dI/dn$
• Collapse criterion: $dI/dn \to 0$ AND $d\Psi/dn \to 0$
• Both conditions must be satisfied simultaneously
• Trajectory analysis across variable space

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MATHEMATICAL FRAMEWORK
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Collapse Potential:
$$\Psi(n,\rho,D) = \Gamma - \Gamma_{crit} - \alpha(I_{avg} - I_{sat})^2 - \beta/S_D$$

Information Saturation:
$$I_{avg} = I_{sat}[1 - \exp(-\lambda(n - n_0))]$$
$$I_{sat} = a + b\rho$$

Coverage Constriction (3D):
$$\Gamma = \Gamma_{max}\cdot\exp(-(n-n_{peak})^2/2\sigma^2)\cdot(1 + \beta(n-n_{peak}))$$

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PARAMETERS
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3D Parameters:
$a = -4.506$, $b = 20.809$
$\lambda = 0.896$, $n_0 = 4.432$

4D Parameters:
$a = -10.899$, $b = 45.211$
$\lambda = 0.292$, $n_0 = -0.172$

Collapse Parameters:
$\alpha = 0.1$, $\beta = 0.5$
$\sigma_{max} = 0.5$

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ANALYSIS SCOPE
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Dimensions: 3D (n=5-16), 4D (n=11-18)
Densities: $\rho \in \{0.3, 0.5, 0.7, 0.9\}$

Simulations:
• Flow field analysis (2 pages)
• Collapse trajectories (8 pages, 4 densities × 2 dimensions)

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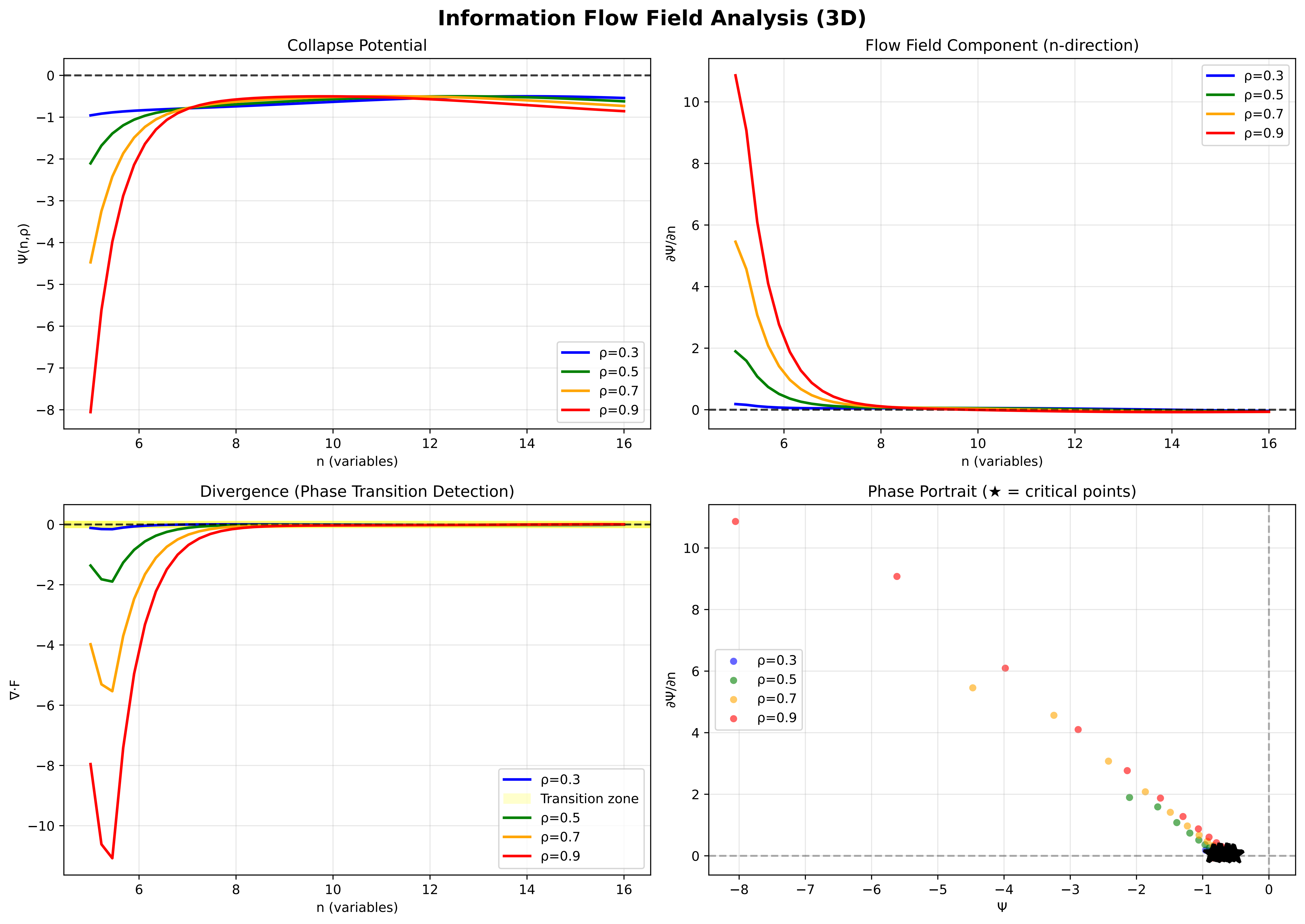
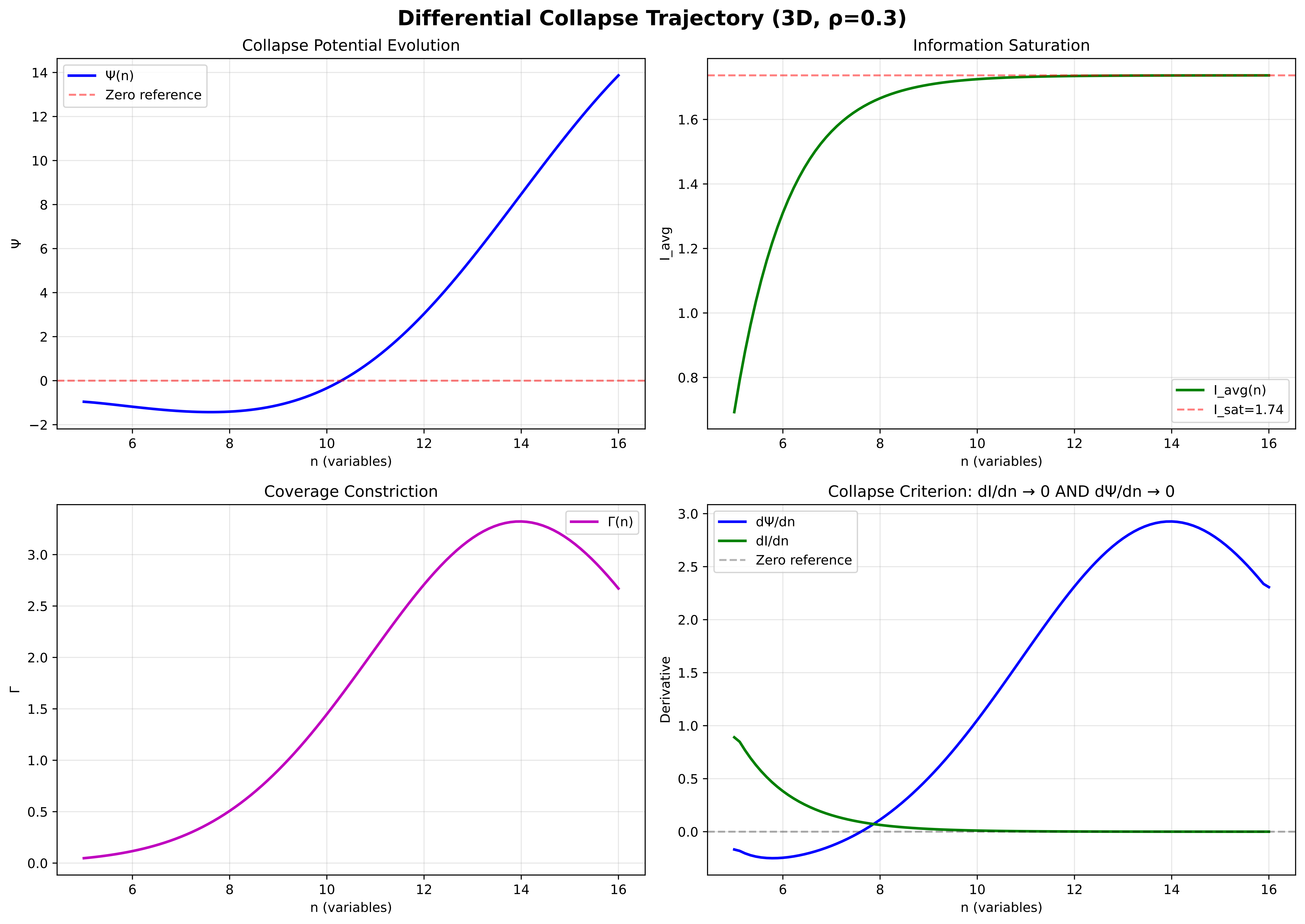
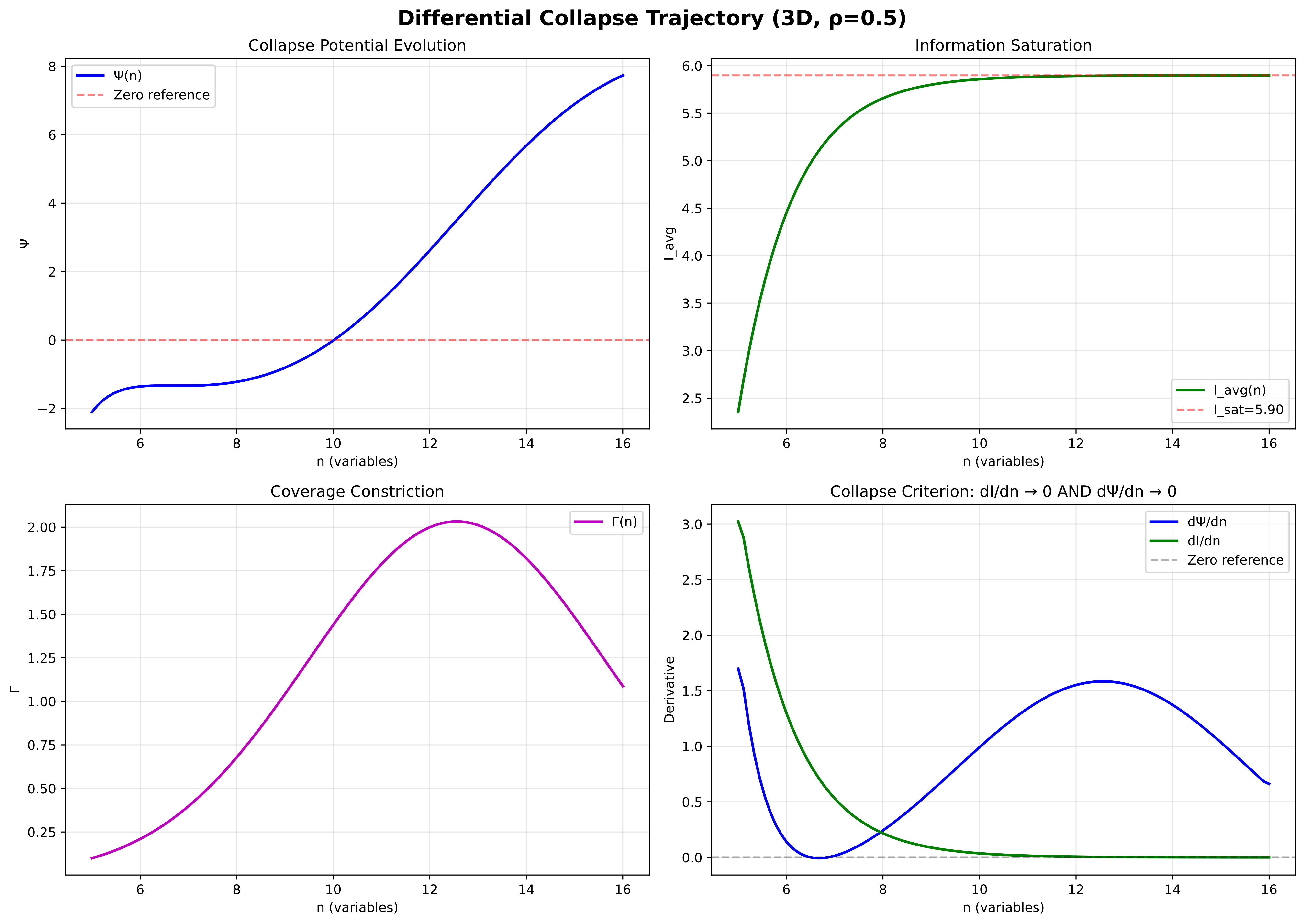
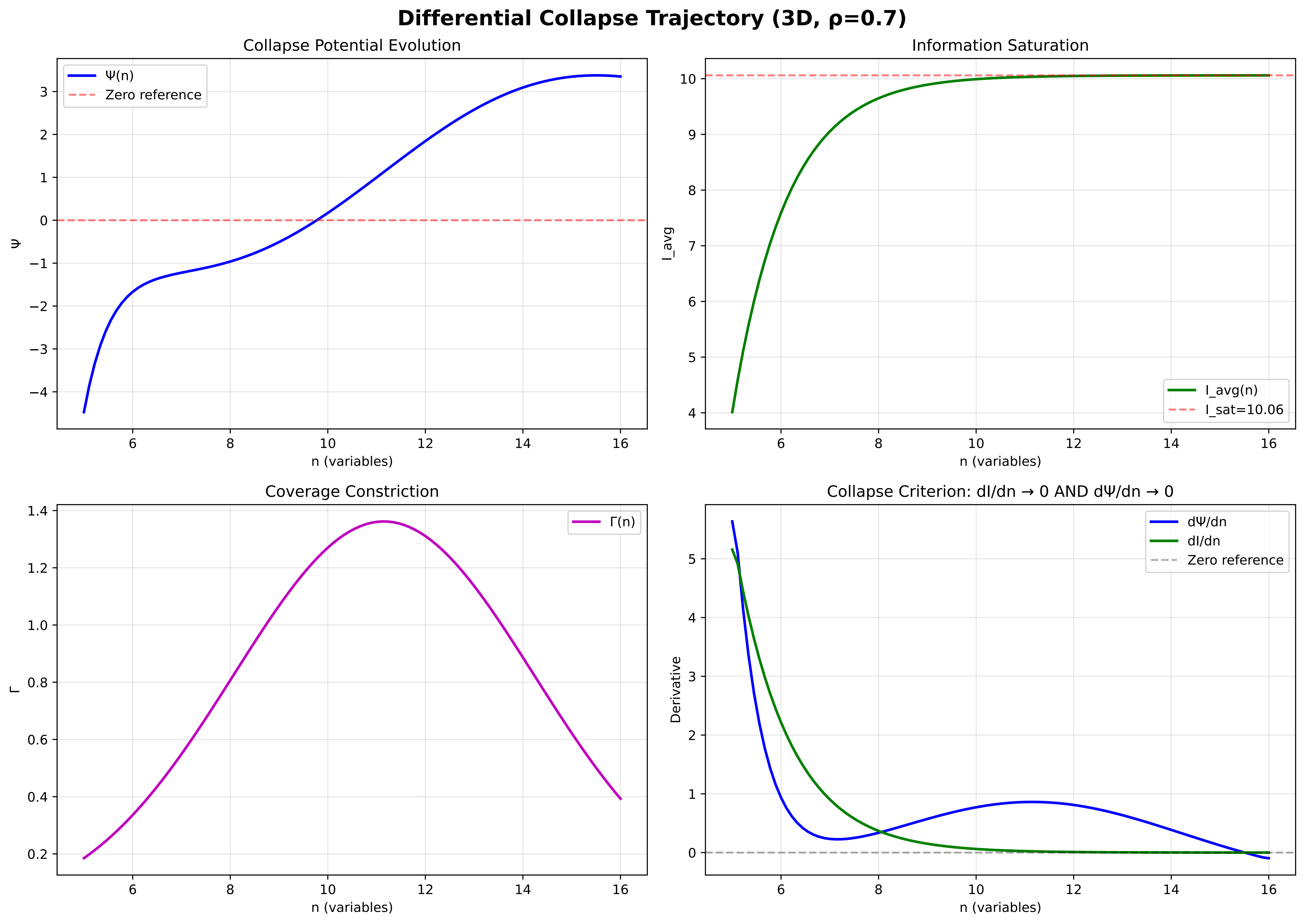

Information Flow Field Analysis (3D)
Collapse Potential
Ψ(n,ρ)
n (variables)
ρ=0.3
ρ=0.5
ρ=0.7
ρ=0.9
Flow Field Component (n-direction)
∂Ψ/∂n
n (variables)
ρ=0.3
ρ=0.5
ρ=0.7
ρ=0.9
Divergence (Phase Transition Detection)
∇·F
n (variables)
ρ=0.3
Transition zone
ρ=0.5
ρ=0.7
ρ=0.9
Phase Portrait (★ = critical points)
∂Ψ/∂n
Ψ
ρ=0.3
ρ=0.5
ρ=0.7
ρ=0.9

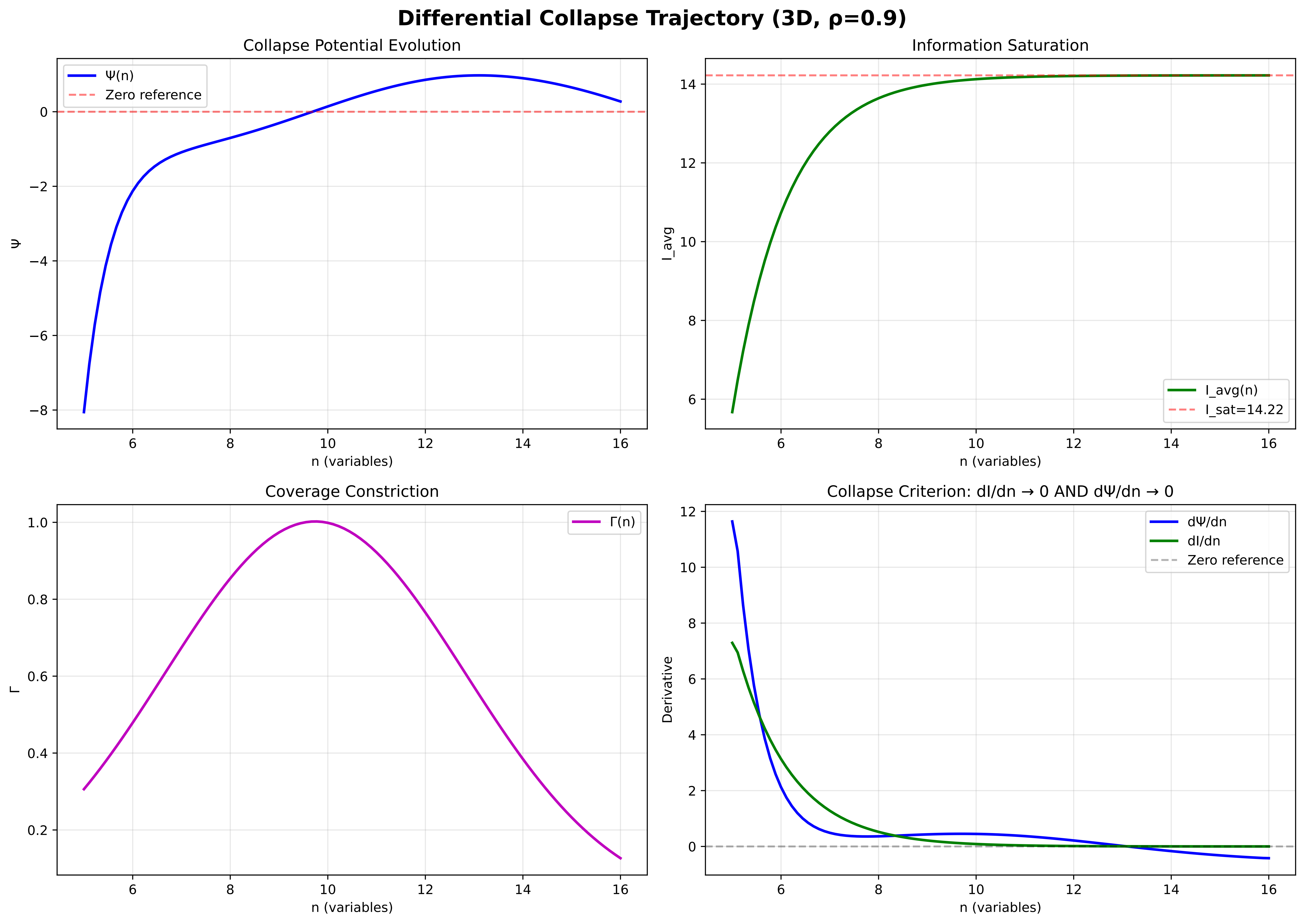


Differential Collapse Trajectory (3D, ρ=0.3)
Collapse Potential Evolution
Ψ(n)
Zero reference
Ψ
n (variables)
Information Saturation
I_avg(n)
I_sat=1.74
I_avg
n (variables)
Coverage Constriction
Γ(n)
Γ
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
dΨ/dn
dI/dn
Zero reference
Derivative
n (variables)

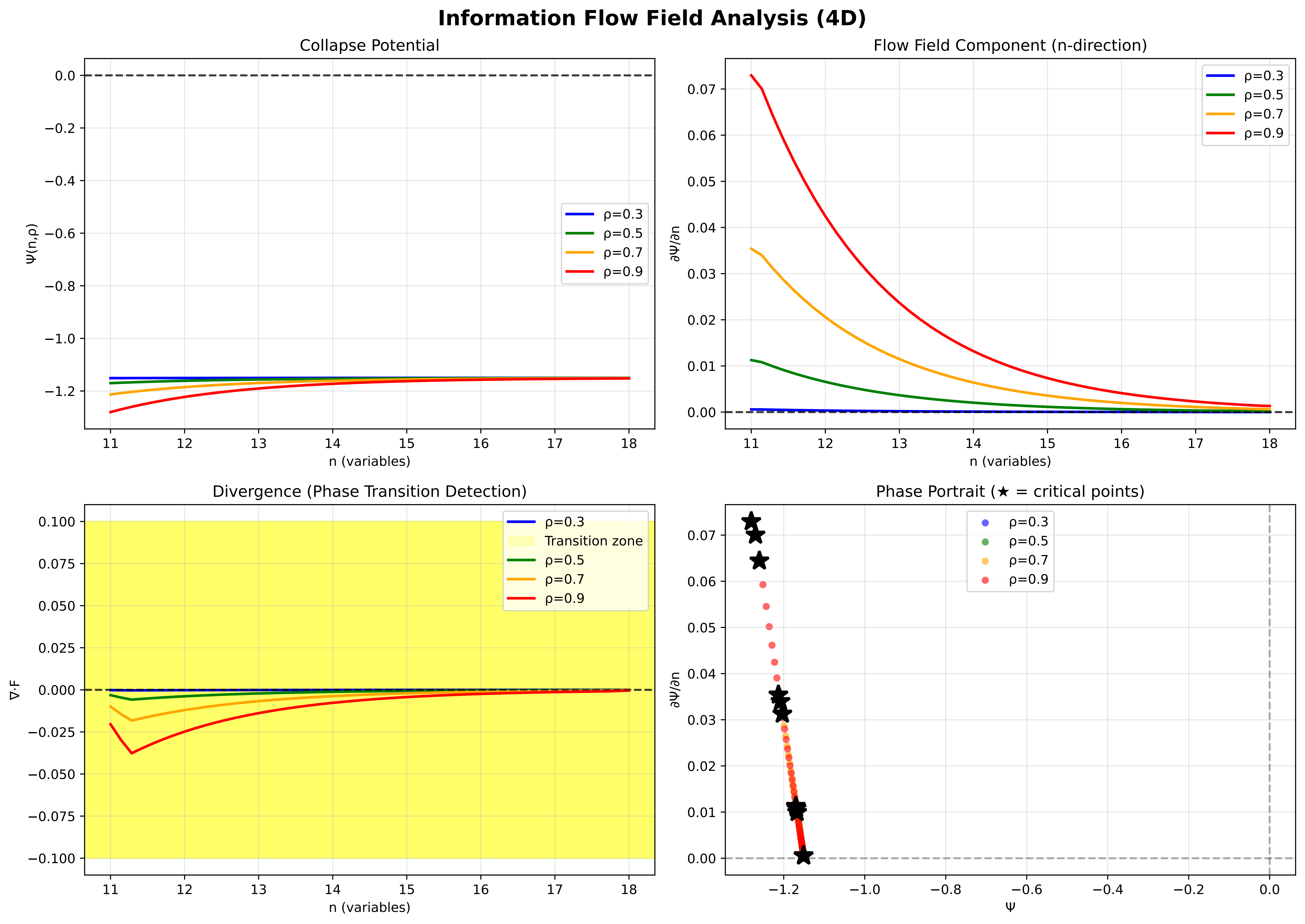

Differential Collapse Trajectory (3D, ρ=0.5)
Collapse Potential Evolution
Ψ(n)
Zero reference
Ψ
n (variables)
Information Saturation
I_avg(n)
I_sat=5.90
I_avg
n (variables)
Coverage Constriction
Γ(n)
Γ
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
dΨ/dn
dI/dn
Zero reference
Derivative
n (variables)



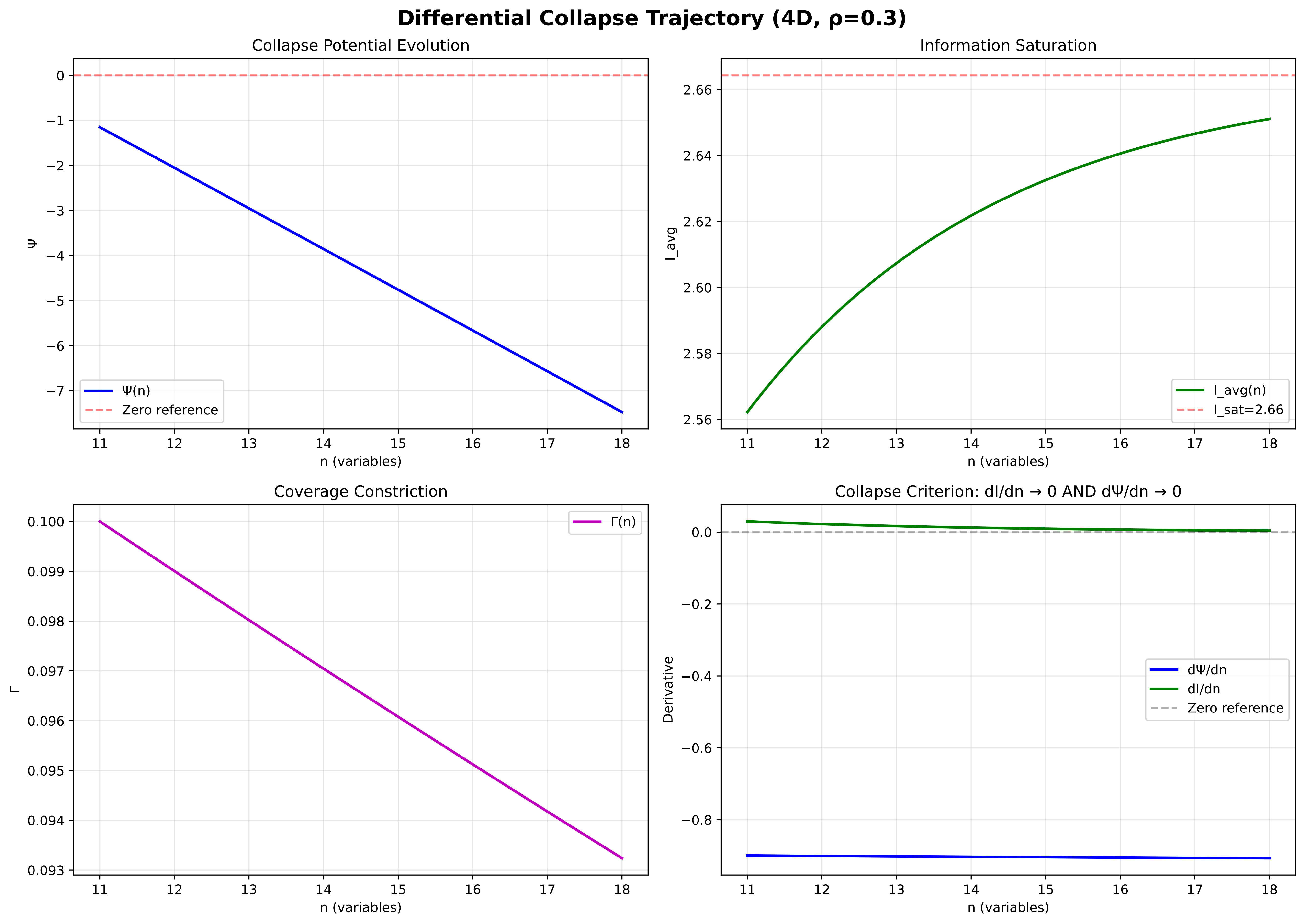
Differential Collapse Trajectory (3D, ρ=0.7)
Collapse Potential Evolution
Ψ(n)
Zero reference
Ψ
n (variables)
Information Saturation
I_avg
I_avg(n)
I_sat=10.06
n (variables)
Coverage Constriction
Γ(n)
Γ
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
dΨ/dn
dI/dn
Zero reference
Derivative
n (variables)

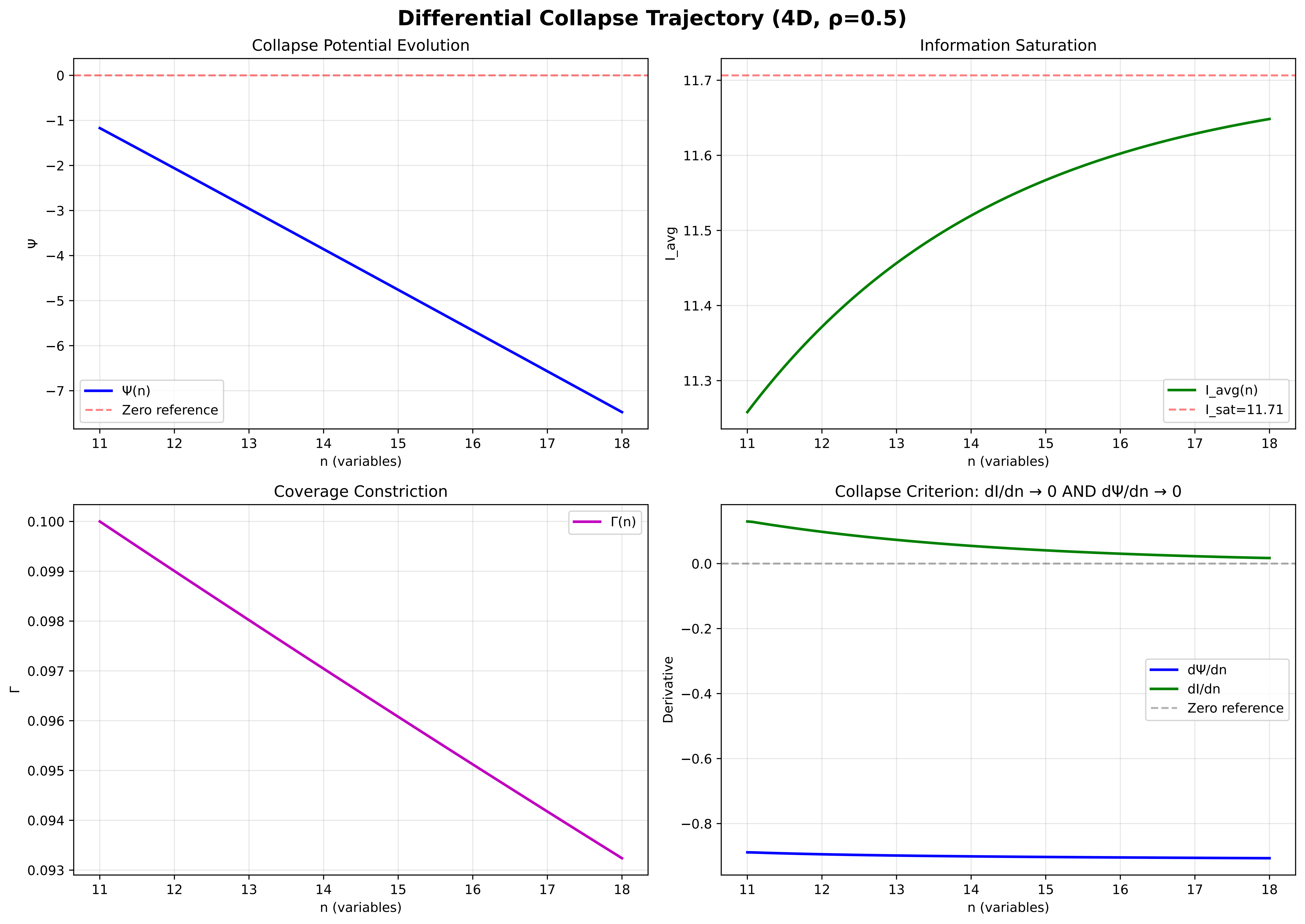


Differential Collapse Trajectory (3D, ρ=0.9)
Collapse Potential Evolution
Ψ(n)
Zero reference
Ψ
n (variables)
Information Saturation
I_avg(n)
I_sat=14.22
I_avg
n (variables)
Coverage Constriction
Γ(n)
Γ
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
dΨ/dn
dI/dn
Zero reference
Derivative
n (variables)

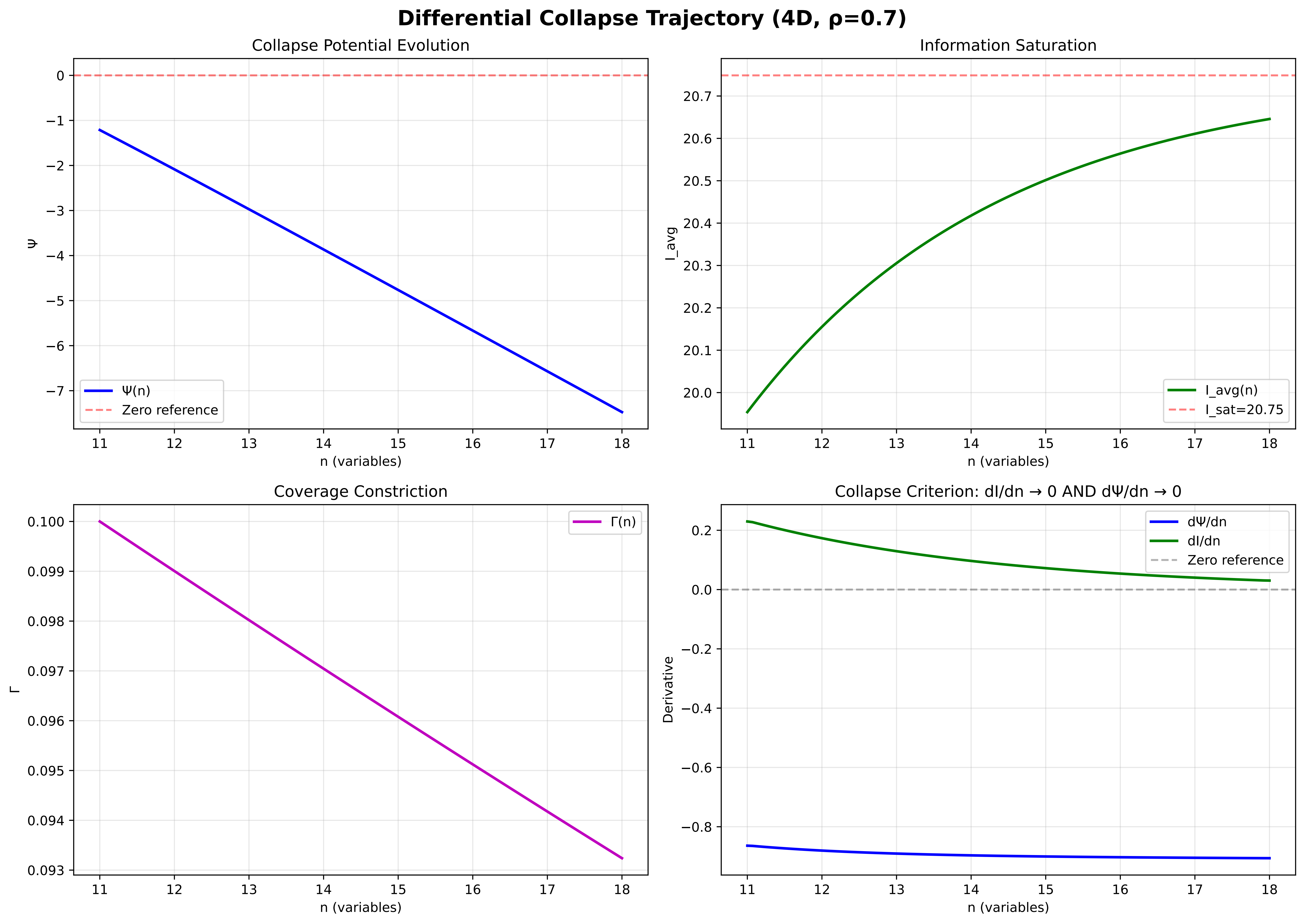


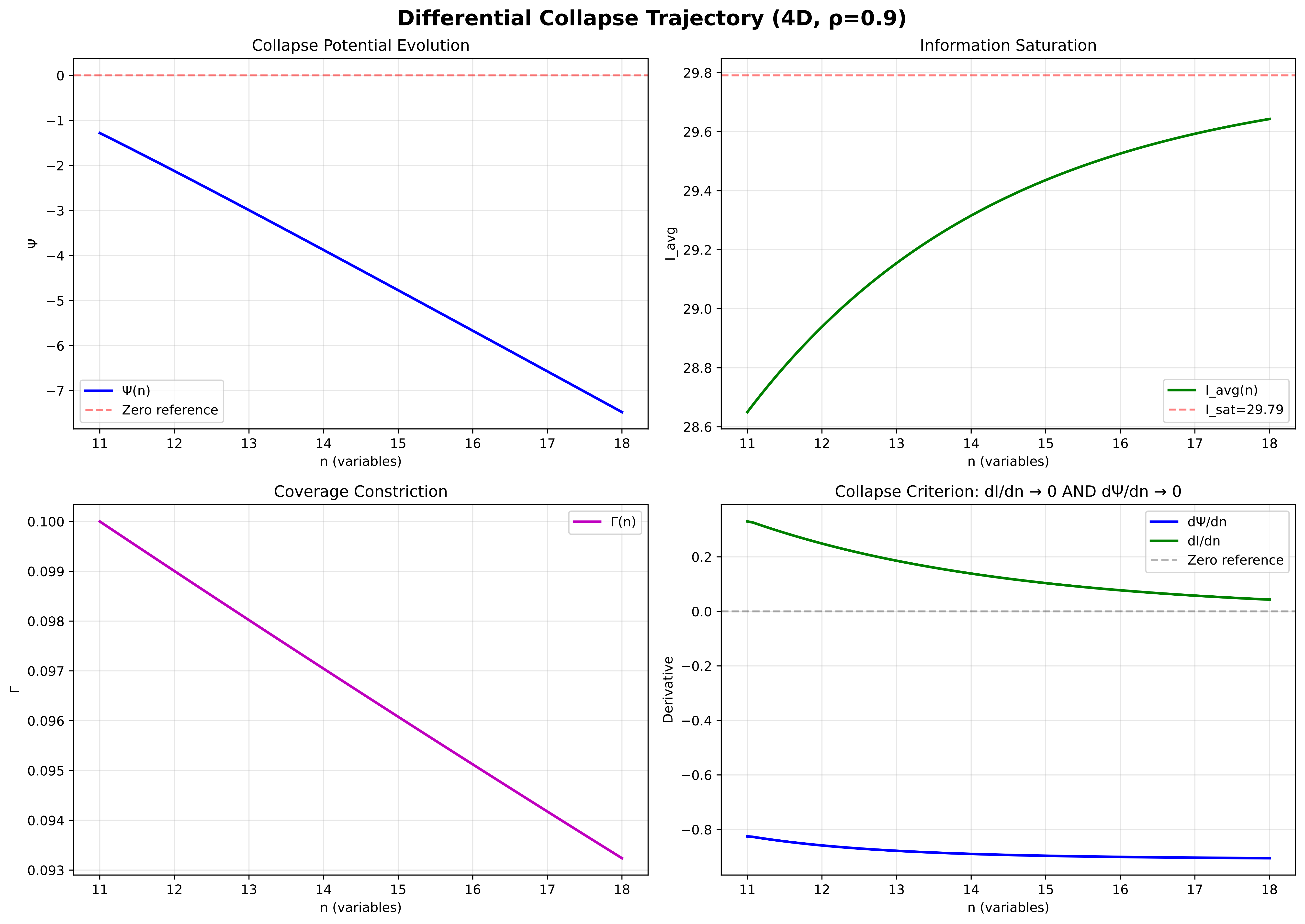
Information Flow Field Analysis (4D)
Collapse Potential
Ψ(n,ρ)
n (variables)
ρ=0.3
ρ=0.5
ρ=0.7
ρ=0.9
Flow Field Component (n-direction)
∂Ψ/∂n
n (variables)
ρ=0.3
ρ=0.5
ρ=0.7
ρ=0.9
Divergence (Phase Transition Detection)
∇·F
n (variables)
ρ=0.3
Transition zone
ρ=0.5
ρ=0.7
ρ=0.9
Phase Portrait (★ = critical points)
∂Ψ/∂n
Ψ
ρ=0.3
ρ=0.5
ρ=0.7
ρ=0.9


Differential Collapse Trajectory (4D, ρ=0.3)
Collapse Potential Evolution
Ψ
Ψ(n)
Zero reference
n (variables)
Information Saturation
I_avg
I_avg(n)
I_sat=2.66
n (variables)
Coverage Constriction
Γ
Γ(n)
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
Derivative
dΨ/dn
dI/dn
Zero reference
n (variables)



Differential Collapse Trajectory (4D, ρ=0.5)
Collapse Potential Evolution
Ψ
Ψ(n)
Zero reference
n (variables)
Information Saturation
I_avg
I_avg(n)
I_sat=11.71
n (variables)
Coverage Constriction
Γ
Γ(n)
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
Derivative
dΨ/dn
dI/dn
Zero reference
n (variables)


Differential Collapse Trajectory (4D, ρ=0.7)
Collapse Potential Evolution
Ψ
Ψ(n)
Zero reference
n (variables)
Information Saturation
I_avg
I_avg(n)
I_sat=20.75
n (variables)
Coverage Constriction
Γ
Γ(n)
n (variables)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
Derivative
dΨ/dn
dI/dn
Zero reference
n (variables)



Differential Collapse Trajectory (4D, ρ=0.9)
Collapse Potential Evolution
Ψ
n (variables)
Ψ(n)
Zero reference
Information Saturation
I_avg
n (variables)
I_avg(n)
I_sat=29.79
Coverage Constriction
Γ
n (variables)
Γ(n)
Collapse Criterion: dI/dn → 0 AND dΨ/dn → 0
Derivative
n (variables)
dΨ/dn
dI/dn
Zero reference